KERALA AGRICULTURAL PRODUCE MARKET COMMITTEES (APMC) DATA ANALYSIS

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AGENDA

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INTRODUCTION

Agricultural Produce Market Committees (APMCs) play a key role in agricultural trade by ensuring fair pricing and market access for farmers. The Kerala APMC Market Analytics 2025 Dashboard provides insights into commodity arrivals, modal prices, market value, and district-wise market trends. These analyses support data-driven decision-making to improve market efficiency, stability, and agricultural growth across Kerala.

OBJECTIVES

01 To analyze the volume of agricultural commodity arrivals across Kerala APMC markets.

02 To evaluate market performance using key indicators such as market value and modal prices.

03 To identify trends and relationships between commodity supply and price fluctuations.

04 To assess district-wise market volatility and stability.

05 To provide actionable insights that support data-driven decision-making for agricultural market management and policy planning.

DATA CONTENTS

- Date
- State
- State Code
- District
- District Code
- Market center
- Market Center Code
- Latitude
- Longitude
- Commodity Type
- Commodity
- Variety
- Origin
- Arrivals
- Arrivals Unit
- Maximum Price
- Minimum Price
- Modal Price
- Price Unit

DATA CLEANING AND PREPROCESSING

- Removed duplicate records.
- Corrected inconsistent district and market center names (e.g., merged duplicate entries such as Kozhikode).
- Handled missing and invalid values.
- Converted data types (dates, prices, arrivals, and quantities) into appropriate formats.
- Filtered the dataset to include only records from Kerala.
- Selected data corresponding to the year 2025 for focused analysis.

KEY MEASURES CREATED

a. Market Value:

Represents the total economic value of commodities traded in markets and reflects overall market activity.

Market Value =∑(Arrivals × Modal Price)

b. Price Spread:

Measures price variability by calculating the difference between maximum and minimum prices. Lower values indicate stable markets, while higher values suggest greater volatility.

Price Spread= ∑(Maximum Price−Minimum Price) / N

where N is the number of records.

KEY MEASURES CREATED

c. Target Spread:

A benchmark used to evaluate market price variability by comparing actual price spread with acceptable limits.

•**Price Spread ≤ Target Spread** → Market prices are within the acceptable variation range.

•**Price Spread > Target Spread** → Indicates higher than desired price fluctuations and potential market instability.

d. Volatility (%):

Measures price fluctuations relative to the average modal price. Higher values indicate greater market instability.

Volatility (%) = (Price Spread / Average Modal Price) × 100

KEY MEASURES CREATED

e. Volatility Target:

A benchmark used to assess market stability by comparing actual volatility with acceptable limits. Lower volatility relative to the target indicates a more stable market.

- **Volatility(%) ≤ Target Volatility** → Market is considered relatively stable.
- **Volatility(%) > Target Volatility** → Market exhibits higher than acceptable price fluctuations and may require monitoring or intervention.

KEY FINDINGS

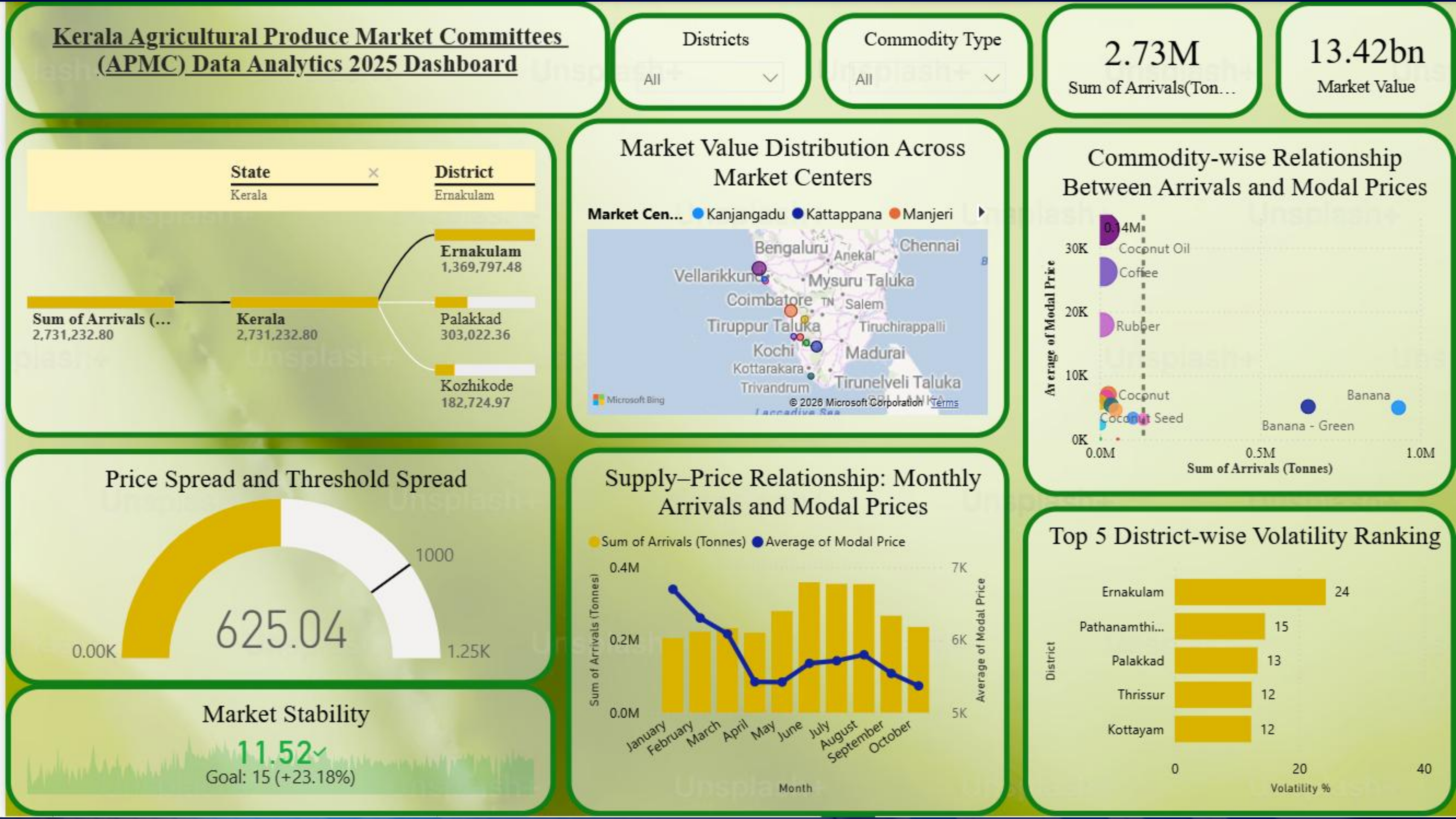
- Kerala APMC markets handled **2.73 million tonnes** of produce worth **₹13.42 billion**.
- **Banana and Banana Green** are the most dominant commodities by arrivals.
- Market prices generally **decrease when arrivals increase**.
- The overall **Price Spread (625.04)** indicates a **moderately stable market**.
- **Ernakulam** shows the highest volatility and market risk among districts.

RECOMMENDATIONS

- Improve storage and transportation facilities in major market centers.
- Monitor and stabilize prices in highly volatile districts like Ernakulam.
- Provide farmers with real-time market price information.
- Promote high-value commodities to increase farmer income.
- Use seasonal trends and analytics for better market planning and decision-making.

CONCLUSION

The Kerala APMC Market Analytics Dashboard provides valuable insights into agricultural market dynamics, commodity performance, price behavior, and market stability across the state. The analysis reveals that market activity is driven by a few dominant commodities, while prices are strongly influenced by seasonal supply patterns. Overall market conditions remain moderately stable, with manageable levels of price volatility. However, certain districts exhibit higher market risk and require focused attention. By leveraging data-driven insights, policymakers, market authorities, and farmers can make informed decisions to improve market efficiency, enhance price stability, and support sustainable agricultural growth in Kerala.



Kerala Agricultural Produce Market Committees (APMC) Data Analytics 2025 Dashboard
Districts
All
Commodity Type
All
2.73M
Sum of Arrivals(Ton…
13.42bn
Market Value
State
Kerala
District
Ernakulam
Sum of Arrivals (…
2,731,232.80
Kerala
2,731,232.80
Ernakulam
1,369,797.48
Palakkad
303,022.36
Kozhikode
182,724.97
Market Value Distribution Across Market Centers
Market Cen… Kanjangadu Kattappana Manjeri
Bengaluru
Anekal
Chennai
Vellarikkundu
Mysuru Taluka
Coimbatore
Salem
Tiruppur Taluka
Tiruchirappalli
Kochi
Madurai
Kottarakara
Tirunelveli Taluka
Trivandrum
Microsoft Bing
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Commodity-wise Relationship Between Arrivals and Modal Prices
0.14M
Coconut Oil
Coffee
Rubber
Coconut
Coconut Seed
Banana
Banana - Green
30K
20K
10K
0K
Average of Modal Price
0.0M
0.5M
1.0M
Sum of Arrivals (Tonnes)
Price Spread and Threshold Spread
625.04
0.00K
1000
1.25K
Supply–Price Relationship: Monthly Arrivals and Modal Prices
Sum of Arrivals (Tonnes) Average of Modal Price
0.4M
0.2M
0.0M
7K
6K
5K
Sum of Arrivals (Tonnes)
Average of Modal Price
January
February
March
April
May
June
July
August
September
October
Month
Top 5 District-wise Volatility Ranking
Ernakulam 24
Pathanamthi… 15
Palakkad 13
Thrissur 12
Kottayam 12
District
0
20
40
Volatility %
Market Stability
11.52
Goal: 15 (+23.18%)

THANK YOU